

OBJECTIVES

- Creating interactive stateful widgets
- Learn how to navigate between screens.

TASKS BEFORE THE LAB

- Learn how to add an icon for the Tic Tac Toe Game
- Students should install and run the Navigation Simple App provided in the lecture.

STRICTLY INDIVIDUAL EXERCISES (NOT TEAMS)

Task / Exercise C1 (Simple Interactive App) [45 mins + 15 mins class demonstration of a working solution]:

- Write a flutter App to reverse a String. A text field is shown to the user to enter a string and a button to click “Reverse String”. The resulting reversed string should be shown at the bottom of the screen.
- Remove the button and let the resulting reversed string be shown immediately upon typing in the text field.
- Show a message at the bottom indicating when the given string is a palindrome. If not, don't show the message.

Task / Exercise C2 (Interactive App - Part 1) [45 mins + 15 minute demonstration of a working solution] :

Warning : This is a single screen exercise.

Continue from the exercise of the previous lab to do the following :

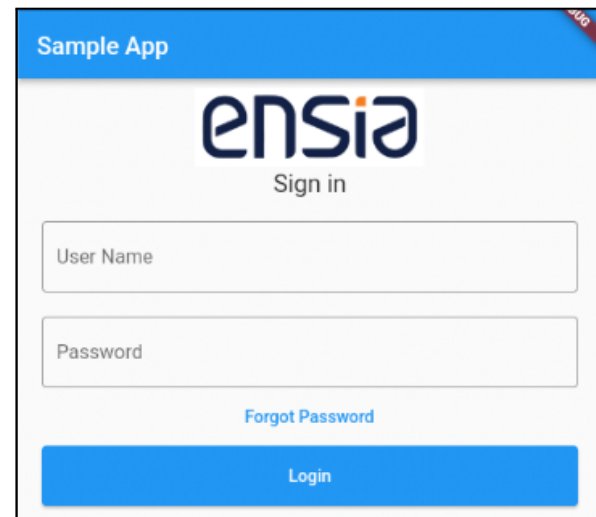
- Put the ENSIA logo at the Top.
- Ensure that the main.dart file is minimal

Assuming the users data is stored inside a local variable as:

```
List<Map<string,dynamic>> data=[  
  {'username':'student1','password':'123', 'name':'Ali'},  
  {'username':'student2','password':'103', 'name':'Aya'},  
  {'username':'student3','password':'203', 'name':'Ahmed'},  
  ...  
];
```

implement the following business logic upon clicking Login:

- If they enter the correct credentials(username/password), show them inside the **same screen**
 - a welcome message “Hello **actual_student_name**”
 - A button “Log out” to show the login screen
 - (You need of course to hide the login form)
- If they enter the incorrect information, show them a red message “Incorrect username or password, try again” at the bottom



Task / Exercise C3 (Interactive App - Part 2) [30 mins] :

For the previous implementation of a one screen App:

1. re-code your app for the login screen in the previous exercise to navigate through **different screens**.
2. Every screen should be written in a separate dart file:
 - a. login_home.dart
 - b. welcome_inside.dart

Task / Exercise C4 (Design & Build Static Screens for your Project) [30 mins] :

Students should collaborate where **each one individually** should work on the design (even on pen and paper) and implementation of a single screen for their project Apps.

OPTIONAL EXERCISES

Task / Exercise P1

Complete the Tic Tac Toe game to add the functionality:

- Detect when there is a winner
- + When there is a draw

Task / Exercise P2

In practice, when we click a login button, the login process can take 5 seconds to authenticate the user, during this time we show the user a **CircularProgressIndicator**.

Extend the code where after clicking “login”, a **circularProgressIndicator** is shown for 5 seconds and later, the business logic would continue as previously.

Task / Exercise P3 :

For the C4, host your work on Firebase hosting service so that it can be accessed online using a browser

Task / Exercise P4 :

For the ToDo App, within a single screen, use sliding pages or tabbed pages to show “To do” items in progress whilst another page (not screen) to show only completed items.

Task / Exercise P5 :

Most students in the year 2023–2024, managed to get an AI-player for Tic Tac Toe. Can students of this year extend the game to play against a computer ?

Task / Exercise P6 :

Additionally, students from previous years managed to create the Snake Game using Kotlin + Flutter in a very short time. Can you do it using Flutter ?